SPONTANEOUS EMISSION BETWEEN MIRRORS*

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The modification of fluorescence lifetimes due to the presence of mirrors is discussed using an image method. The close analogy of this system with that of many-atom cooperative decay is exploited and discussed.

Spontaneous emission of radiation has the interesting property of being affec ed by the source atom's local environment. Such an environmental influence might be, for example, the presence of neighboring atoms which couple by their radiative interactions to generate the phenomenon of cooperative decay [1, 2]. (A classic example of the way the environment modifies the radiative properties is the Lorentz local field in dielectrics.) Recently several authors have studied the related effect of placing the source atom between infinite plane parallel mirrors [3–6]. The approach used is straightforward: the electromagnetic field is expanded in appropriate mode functions satisfying the boundary conditions imposed by the mirrors. An alternative approach, alluded to in [6], is the image method which replaces the mirror cavity by an infinite string of virtual images. Stehle [3] has used such a method for the case where the mirrors have finite conductivity, but it does not appear that the perfect conductivity results of refs. [4–6] can be easily recovered. In this note we use the image method for perfectly conducting mirrors to derive the result obtained by the mode expansion approach. We also comment on the connection between this result and the cooperative effects noted above.

The image concept has been used by Morawitz [7] in a semiclassical treatment of the simpler problem of an excited two-level atom a distance Z away from a perfectly reflecting plane mirror. Since this is the limiting case of the two-mirror calculation as the distance Z of the atom from one mirror is held fixed and the mirror spacing L is taken to infinity, we find it convenient to discuss this a little further. The system behaves as a Lloyd's mirror: we have the possibility of interference between the two processes shown in fig. 1a and 1b. In 1a, the atom emits into the mode $|\boldsymbol{k}_\lambda\rangle$ ($\boldsymbol{k}$ is the wave-vector, λ the polarisation of the plane wave mode), without reflection from the mirror. Alternatively a photon can be emitted into mode $|\boldsymbol{k}_\lambda\rangle$ by reflection (fig. 1b) and can be considered to be emitted by a fictitious *image* atom. The photon carries no information as to which process has occurred.

Thus we may describe the emission as a two-atom problem with complete uncertainty as to which atom is excited. The initial states incorporating such uncertainty are

$$|\Psi_\pm\rangle = (1/\sqrt{2})\,\{|+\rangle_A\,|-\rangle_I \pm |-\rangle_A\,|+\rangle_I\}\,, \tag{1}$$

where A (I) refers to the atom (image) and $|+\rangle$ ($|-\rangle$) to the excited (ground) state. The atom–image separation is $2Z$. State $|\Psi_+\rangle$ is used if the atomic transition dipole is normal to the plane of the mirror: the dipole then has the same phase as its image. Similarly state $|\Psi_-\rangle$ is used if the transition dipole is parallel to the plane of the mirror, and represents a dipole with the phase opposite to that of the image. This then corresponds to the classical description of an oscillating dipole and its image. The spontaneous emission lifetime appropriate to the initial state (1)

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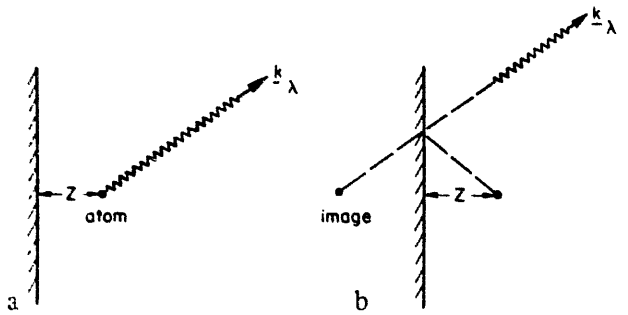


Fig. 1. Spontaneous emission processes for an atom in front of one mirror.

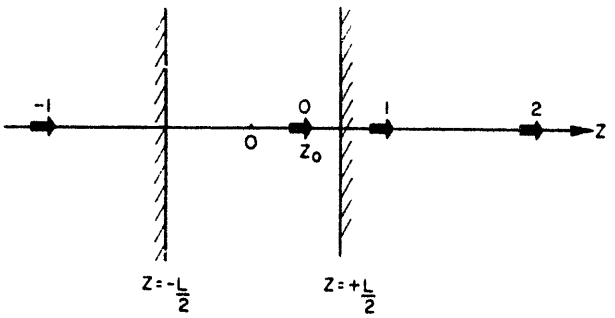


Fig. 2. An atom and its associated images from two mirrors.

may be inferred from the work of Stephen [2]; Morawitz has shown that the results, when applied to an atom in front of a mirror, agree well with experiment [8]. These are the same results reported recently by Philpott [6]. The cooperative state (1) implies Dicke superradiance for $|\Psi_+\rangle$ (the lifetime is halved) as $Z \to 0$, and a subradiance for $|\Psi_-\rangle$ (where the system does not decay at all) as $Z \to 0$.

Effects of retardation may be included in the atom–image interaction: two atoms in the state $|\Psi_\pm\rangle$ at $t = 0$ and separated by a distance $2Z$ decay independently until time $t = 2Z/c$ [9]. In our problem, this implies that the atom decays as though in free space until it "sees" the mirror.

Now consider the problem of an excited atom between two perfectly conducting mirrors located at $Z = \pm \frac{1}{2}L$ (fig. 2), for the case where the atomic transition dipole moment is oriented along the Z axis. Using the image method, this is equivalent to an infinite chain of atoms in free space, with an initial state

$$|\Psi\rangle = \lim_{N\to\infty} \frac{1}{\sqrt{N}} \sum_{p=1}^{N} |\phi_p\rangle = \lim_{N\to\infty} |\Psi_N\rangle . \tag{2}$$

The state $|\phi_p\rangle$ is one in which atom p is excited and all others are in their ground states. In Dicke's notation [1], $|\Psi_N\rangle$ is a state with $r = \frac{1}{2}N$, $m = 1 - \frac{1}{2}N$. All atoms lie on the Z axis with the position of the pth atom given by

$$Z_p = pL + (-1)^p Z_0 \tag{3}$$

(the origin of coordinates $Z = 0$ is midway between the mirrors, and Z_0 is the position of the real atom as in fig. 2).

The spontaneous emission rate may be calculated using the standard Weisskopf–Wigner approximation [10] and is given by twice the real part of the complex self-energy

$$\Delta^{\perp}(\epsilon) = -i \sum_{\boldsymbol{k}_\lambda} \frac{|g_{\boldsymbol{k}_\lambda}|^2}{\omega_{\boldsymbol{k}} - \omega_0 - i\epsilon} \{1 + \exp[2ik_Z (Z_0 - \tfrac{1}{2}L)]\} \sum_{n=-\infty}^{\infty} \exp(2ink_Z L), \tag{4}$$

where ϵ is a small positive number, ω_0 the transition frequency of the two-level atom and $k_Z \equiv \boldsymbol{k} \cdot \hat{\boldsymbol{Z}}$. We have defined

$$|g_{\boldsymbol{k}_\lambda}|^2 = (\mu^2/\hbar^2)\, |\hat{\boldsymbol{\mu}} \cdot \hat{\boldsymbol{\epsilon}}_{\boldsymbol{k}_\lambda}|^2 (2\pi\hbar\omega_{\boldsymbol{k}}/V) , \tag{5}$$

where $\boldsymbol{\mu} = \mu\hat{\boldsymbol{\mu}}$ is the transition dipole moment, $\hat{\boldsymbol{\epsilon}}_{\boldsymbol{k}_\lambda}$ a polarisation unit vector for mode $|\boldsymbol{k}_\lambda\rangle$ ($\boldsymbol{k} \cdot \hat{\boldsymbol{\epsilon}}_{\boldsymbol{k}_\lambda} = 0$, $\lambda = 1, 2$) and V the quantisation volume. Using the Poisson sum rule

$$\sum_{n=-\infty}^{\infty} \exp(2ink_Z L) = \sum_{n=-\infty}^{\infty} \delta\left(\frac{k_Z L}{\pi} - n\right) \tag{6}$$

we notice the appearance of the Fabry–Pérot modes. In the continuum limit $V \to \infty$, from (4)

$$2\,\mathrm{Re}\,\Delta^{\perp}(\epsilon) = (6\pi A/k_0 L)\left[\tfrac{1}{2} + \sum_{n=1}^{[k_0 L/\pi]} (1 - n^2\pi^2/k_0^2 L^2)\cos^2(n\pi Z_0/L - \tfrac{1}{2}n\pi)\right], \tag{7}$$

where $k_0 = \omega_0/c$, $[k_0 L/\pi]$ is the greatest integer part of $k_0 L/\pi$, and A is the Einstein A-coefficient for spontaneous emission by a single atom in free space.

For the case when the transition dipole moment is parallel to the mirrors, we have instead of (2), the initial state

$$|\psi^{\parallel}\rangle = \lim_{N\to\infty} (1/\sqrt{N}) \sum_{n=1}^{N} (-1)^p |\phi_p\rangle, \tag{8}$$

and the decay rate is

$$2\,\mathrm{Re}\,\Delta^{\parallel}(\epsilon) = (3\pi A/k_0 L) \sum_{n=1}^{[k_0 L/\pi]} (1 + n^2\pi^2/k_0^2 L^2)\sin^2(n\pi Z_0/L - \tfrac{1}{2}n\pi). \tag{9}$$

The results (7) and (9) obtained by the image method are identical to the mode expansion results of Barton [5] and Philpott [6], and are valid for $L \ll c/A$. The imaginary part of the self-energy Δ corresponds to a modified Lamb shift, discussed at length in [5]; this interesting effect does not appear to have been observed.

The modification of the fluorescence lifetime due to the presence of the two mirrors can be thought of as an effect of cooperative emission from a correlated state of the atom and its associated images. Dicke [1] first emphasized that correlated states of the type we have constructed for the image-atom system lead to such cooperative effects.

A second aspect of such cooperation is the emission into certain preferred directions. Thus in the present calculation we have the appearance of Fabry–Pérot modes in the radiation emitted by an infinite one-dimensional chain of atoms sharing one excitation. This diffraction pattern character is also evident in the emitted radiation from an N-atom uncorrelated Bloch state (see for example Rehler and Eberly [1]).

The experimental observation that fluorescence lifetimes depend upon the atom's surroundings was apparently met with some surprise: how could the spontaneous emission of a photon be affected by the local environment, since the atom can only "see" its surroundings by emitting a photon in the first place? Such confusion can only arise from a misunderstanding of the photon concept. For times less than the transit time from the source atom to the nearest point of environmental influence (a mirror for example) and back again, the atom decays as if isolated. For longer times, we have the possibility of reabsorption and emission and, as we have seen, the influence of the environment becomes important. The radiated field (or the photon "wave function") contains all this information: it does not represent a buckshot or fuzzy ball photon [11].

It is natural to investigate now how an excited atom behaves in a *changing* environment. For example, consider a beam of excited atoms moving past a conducting undulating surface. We would expect to find both a *time-dependent* frequency shift and decay rate. This effect would be reminiscent of the Smith–Purcell effect for charged particles [12]. We are currently studying the application of the image technique to this problem. Fuller details of the analysis of this problem, as well as the two-mirror problem, will be presented elsewhere.

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